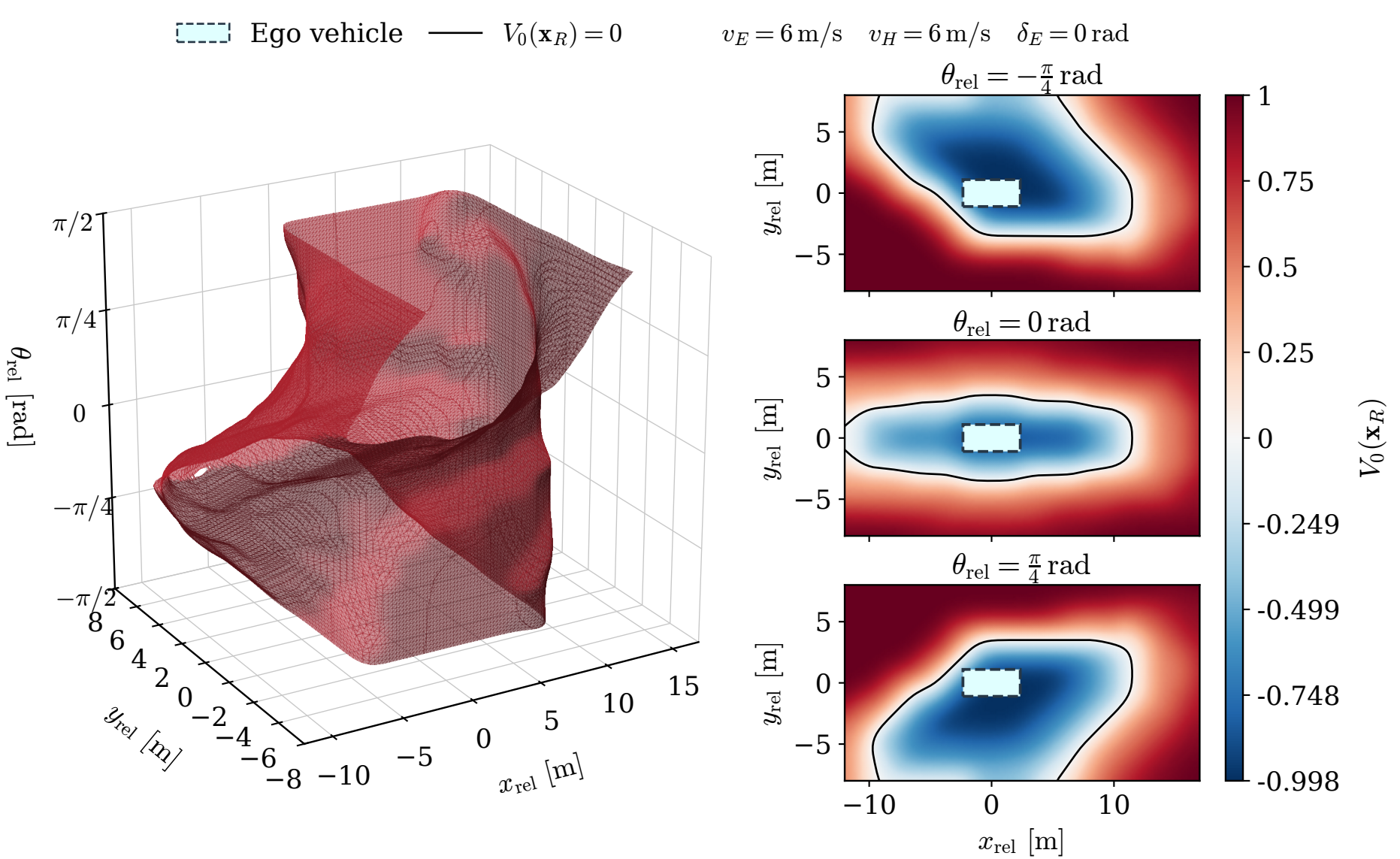

Ego vehicle
$V_0(\mathbf{x}_R) = 0$
$v_E = 6\,\mathrm{m/s}$ $v_H = 6\,\mathrm{m/s}$ $\delta_E = 0\,\mathrm{rad}$
$\theta_{\mathrm{rel}}$ [rad]
$\pi/2$
$\pi/4$
0
$-\pi/4$
$-\pi/2$
$y_{\mathrm{rel}}$ [m]
$x_{\mathrm{rel}}$ [m]
$\theta_{\mathrm{rel}} = -\frac{\pi}{4}$ rad
$\theta_{\mathrm{rel}} = 0$ rad
$\theta_{\mathrm{rel}} = \frac{\pi}{4}$ rad
$V_0(\mathbf{x}_R)$
1
0.75
0.5
0.25
0
-0.249
-0.499
-0.748
-0.998